



Supply Shield

Supply & Fleet Management

Supply Shield

Supply Chain Disruption Assistant
& Fleet Utilisation Optimizer

Team Nexus

AI Track (IBM Bob Hackathon)



Static Logistics Systems **Fail** in Dynamic Crises



3+ Hours Triage Latency

Dispatchers manually cross-reference manifests during port strikes and extreme weather.



\$35B+ Perishable & Pharma Loss

Undetected temperature excursions during transit ruin sensitive cargo.



Idle Capital Imbalance

Vehicles sit idle in safe hubs while neighboring crisis corridors face severe capacity deficits.

Turning Static Tracking into Autonomous Operational Triage



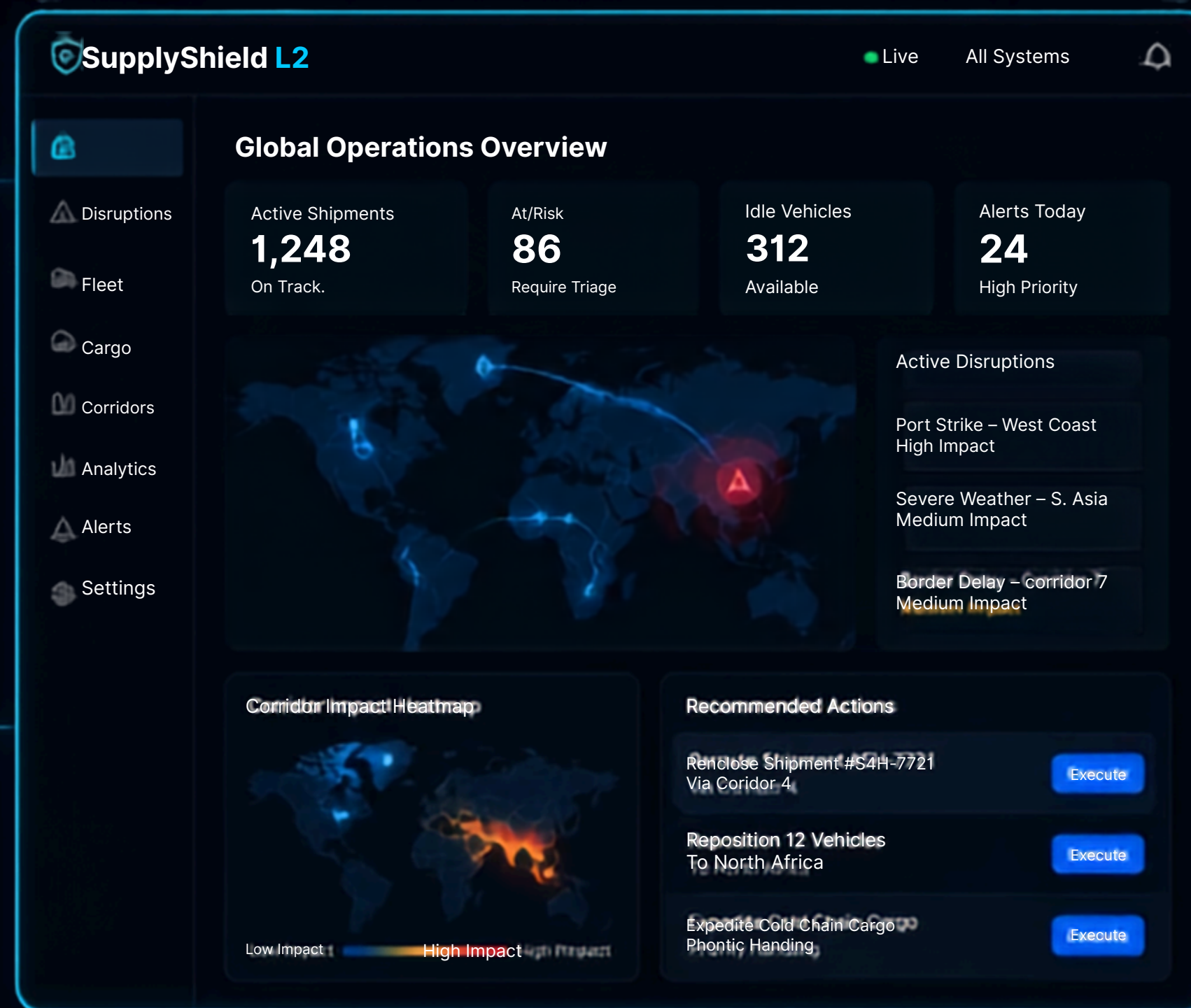
Disruption Command Panel

Corridor impact matching with 1-click refooting.



Fleet Redployment Optimizer

Algorithmic 0–100 scoring
balancing idle hours against
regional demand spikes.



IoT Cold Chain Telemetry

Real-time sensor streaming
with automated WHO PQS /
FDA 21 CFR severity grading.

Load-Bearing IBM Bob AI

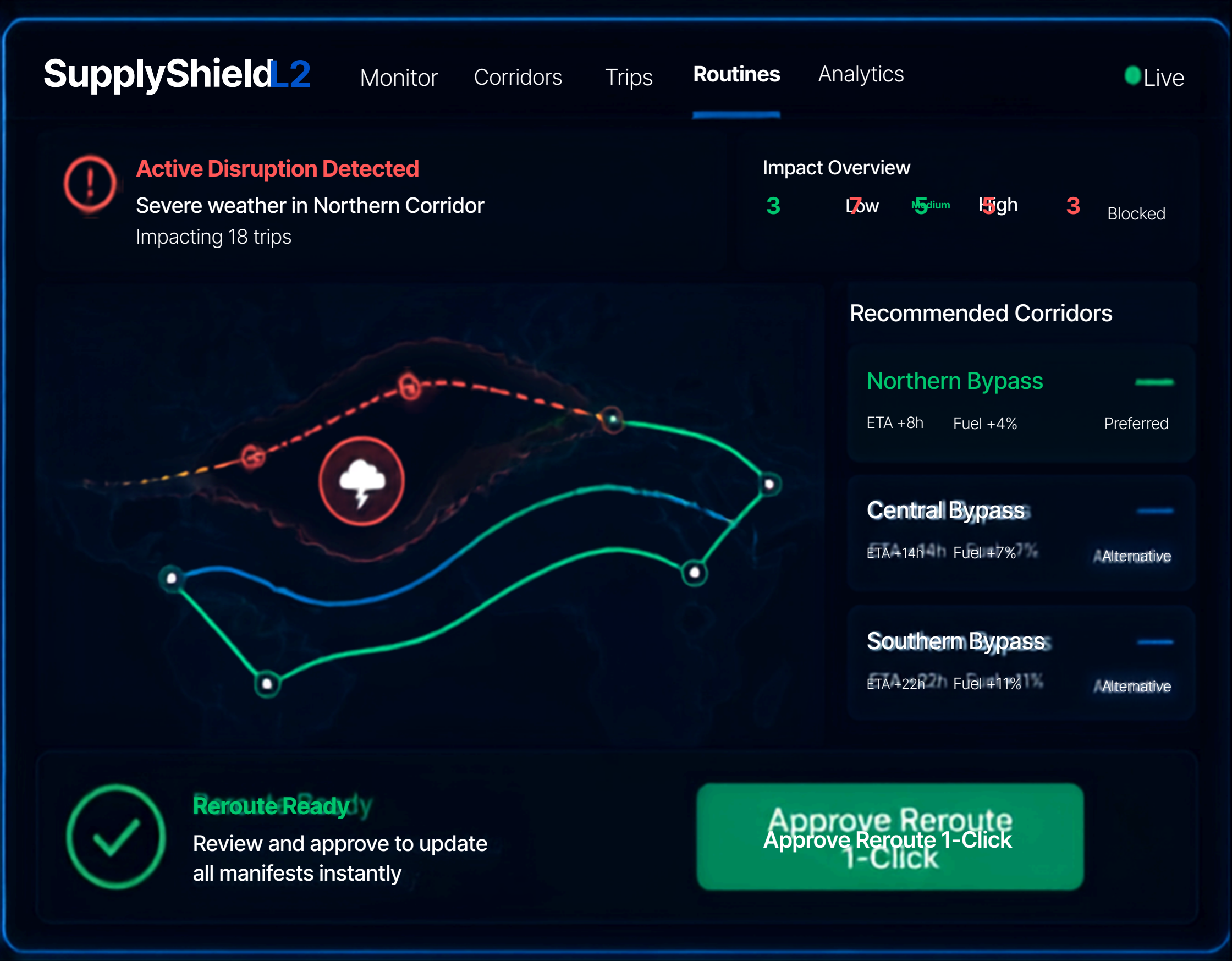
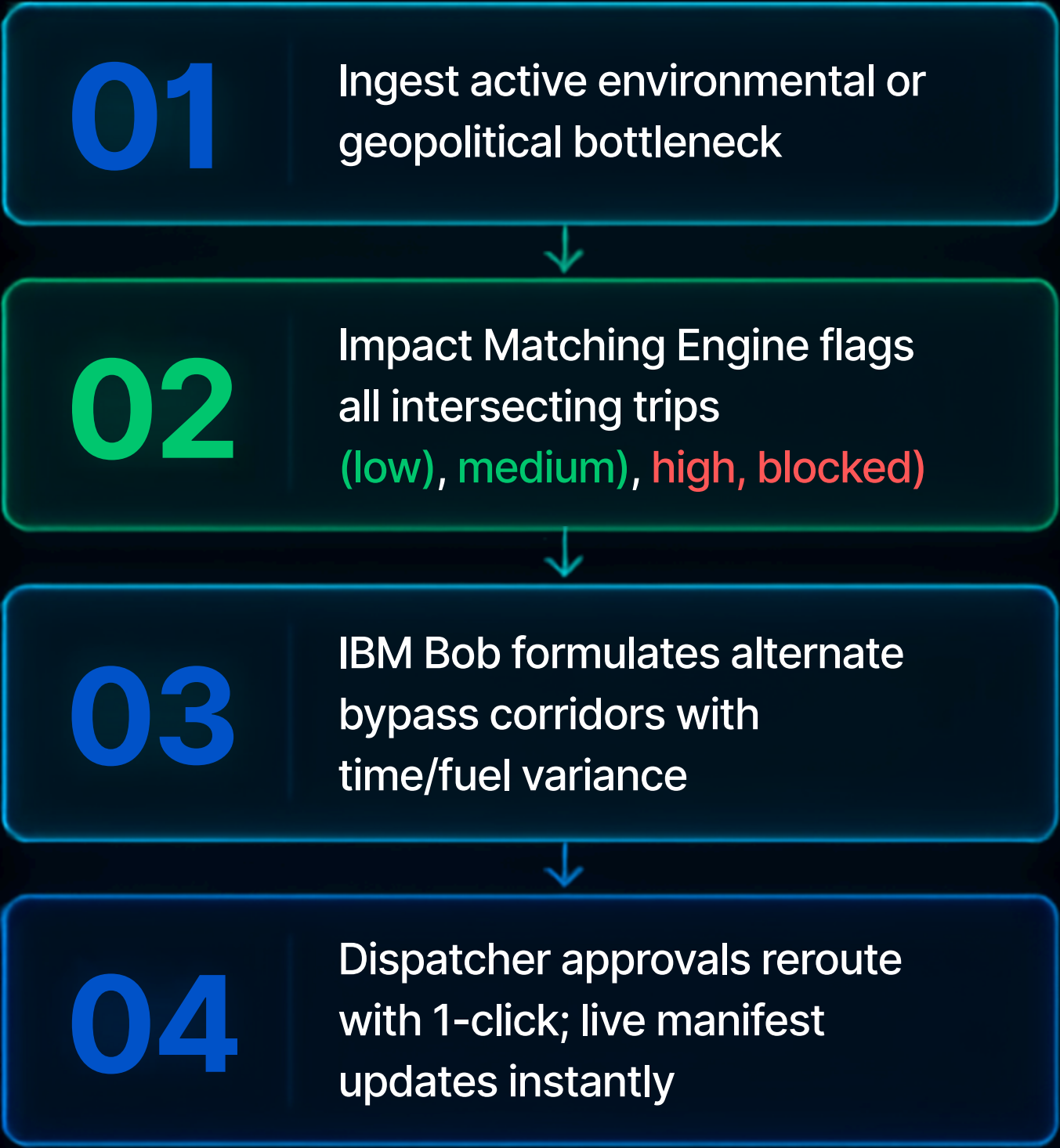
Autonomous agent triage
powered by 13 live Model
Context Protocol (MCP) tools.

Deterministic Logic Grounded in Enterprise PostgreSQL



Zero hallucination, strict 4-role RBAC, and auditable execution traces.

Autonomous Corridor Analysis in Action



13 Live Operational Tools Running on Production Database Records

MCP

Official Model Context
Protocol (MCP) Server
(@modelcontextprotocol/sock)
via stdio JSON-RPC 2.0
and in-app **Ask Bob AI** drawer.

IBM Bob AI

13 TOOLS

13 Autonomous Tools
spanning **Shipment Manifests**,
Corridor Disruption Triage,
Idle Fleet Scoring,
and **Cold Chain Excursion Analysis**.

watsonx.ai

watsonx.ai **Granite 3.0** for operational reasoning
and explainable dispatch rationales.

Quantified Efficiency Gains in Logistics Resilience

80% ↓

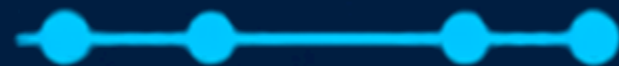
**Reduction in Disruption
Triage Latency**
(from hours to seconds)

ZERO

**Regulatory Cold Chain
Spoilage Breaches**
with proactive alerts

+35% ↑

Fleet Asset Utilization
via demand-driven
idle redeployment



**Roadmap for directMQTT/CoAP edge IoT gateway
ingestion and offline edge buffers.**

Team Nexus & Core Contributions

Built with Passion for the IBM Bob AI Hackathon



Preet Makadia

Team Lead &
AI/ MCP Architect

IBM Bob MCP Server
(13 tools)
system architecture
full-stack integration



Ronak Chhaniyara

Backend &
Database Engineer

Supabase PostgreSQL Schema
migrations
RLS
showcase data pipeline



Jeel Parsaniya

Frontend &
UI/UX Engineer

Next.js 16 Command
Center Disruption Panel
Recharts telemetry
visualizations



Yaman Ladani

IoT & Cold Chain
Specialist

IoT telemetry engine
WHO/FDA excursion
grading rules
QA validation

GITHUB REPOSITORY <https://github.com/PreetMakadia595/bob-ai-hackathon--Team-Nexus->